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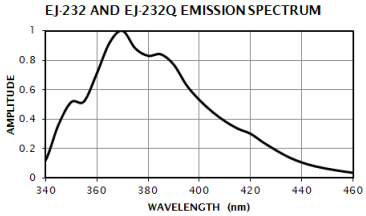
FAST TIMING PLASTIC SCINTILLATOR EJ-232, EJ-232Q

EJ-232 plastic scintillator is intended for very fast timing applications or when very high pulse pair resolution is required. Due to the small emission wavelength, the optical mean free path of this scintillator is approximately 10 cm. Therefore, to achieve the best light collection and to optimize the timing performance, EJ-232 should be used in a small size with the largest scintillator dimension less than 10 cm to minimize photon scattering effects. The use of light guides is best avoided.

EJ-232Q plastic scintillator is a quenched variant of EJ-232 specifically formulated for ultra-fast counting applications. The introduction of small amounts of benzophenone to EJ-232 significantly shortens the timing properties for purposes of achieving very high counting rates or improved coincidence timing. The quenching does not affect the emission spectrum but does reduce the scintillation efficiency (see table below). While it is recommended to keep the scintillator size and shape small in order to achieve the best timing performance, it is not recommended to use EJ-232Q in thin films (thicknesses ≤ 3 mm) due to the vapor pressure of benzophenone.

PROPERTIES	EJ-232	EJ-232Q (% Benzophenone)				
		0.5	1.0	2.0	3.0	5.0
Light Output (% Anthracene)	55	19	11	5	4	3
Scintillation Efficiency (photons/1 MeV e^-)	8,400	2,900	1,700	770	610	460
Wavelength of Maximum Emission (nm)	370	370	370	370	370	370
Rise Time (ps)	350	110	105	100	100	100
Decay Time (ps)	1600	700	700	700	700	700
Pulse Width, FWHM (ps)	1300	360	290	260	240	220
No. of H Atoms per cm^3 ($x10^{22}$)	5.13	5.12	5.12	5.12	5.12	5.12
No. of C Atoms per cm^3 ($x10^{22}$)	4.66	4.66	4.66	4.66	4.66	4.66
No. of Electrons per cm^3 ($x10^{23}$)	3.30	3.38	3.38	3.38	3.38	3.38
Density (g/cm^3)	1.023	1.023	1.023	1.023	1.023	1.023
Polymer Base	Polyvinyltoluene					
Refractive Index	1.58					
Softening Point	75°C					
Vapor Pressure	Vacuum-compatible					
Coefficient of Linear Expansion	7.8 x 10^{-5} below 67°C					
Light Output vs. Temperature	At 60°C, L.O. = 95% of that at 20°C. No change from -60°C to 20°C.					
Temperature Range	-60°C to 60°C					

Chemical Compatibility
Attacked by: Aromatic solvents, Chlorinated solvents, Ketones, Solvent bonding cements, etc.
Stable in: Water, Dilute acids and alkalis, Lower alcohols, and Silicone greases.
It is safe to use most epoxies with this scintillator.



- Plastic Scintillators
  - EJ-200, EJ-204, EJ-208, EJ-212
  - EJ-214
  - EJ-228, EJ-230
  - EJ-232, EJ-232Q
  - EJ-240
  - EJ-244, EJ-248, EJ-244M, EJ-248M
  - EJ-254
  - EJ-256
  - EJ-260, EJ-262
  - EJ-270
  - EJ-276D
  - EJ-290
  - EJ-296
  - EJ-299-33A, EJ-299-34
  - Non-Fogging
- Liquid Scintillators
- Wavelength Shifters
- Fibers
- Light Guides and Acrylic Plastic
- Neutron Detectors
- Zinc Sulfide Coated
- Accessories

Printable PDFs

- EJ-232, EJ-232Q Data Sheet
- EJ-232 SDS
- EJ-232Q SDS

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